

KASIMALLA KRISHNA VAMSI

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SUMMARY

Electrical & Electronics Engineer working at the intersection of robotics hardware and power electronics. Experience spans custom PCB design in KiCad and Altium Designer, ROS 2 autonomy stacks for field robots, and MPPT-based solar power systems. Three papers accepted at IEEE BITCON 2024. Seeking robotics hardware and embedded systems roles where electronics design and autonomy meet.

PROFESSIONAL EXPERIENCE

Robotics Engineer Jun 2025 - Present

Peepul Agri Ventures LLP — Vijayawada, India

- Promoted from intern to lead electrical and electronics development for autonomous agriculture robots, owning hardware from schematic through field deployment.
- Design and route custom PCBs in **KiCad** and **Altium Designer** for power distribution, motor-driver and sensor-interface boards; manage component selection, DFM review and fabrication handoff.
- Contribute to the autonomy software stack, including **ROS 2**-based control, sensor integration and hardware bring-up.

Robotics Intern Dec 2024 - May 2025

Peepul Agri Ventures LLP — Vijayawada, India

- Simulated and tuned swerve-drive control for the agriculture bot in **Ignition Fortress**, validating kinematics before hardware deployment.
- Built and integrated the robot's electronics hardware with a **ROS 2** framework, supporting **PCB** layout and assembly work in **KiCad** and **Altium Designer**.

President, Electric Vehicle Club · Member, Robotics Club Aug 2023 - Apr 2025

Velagapudi Ramakrishna Siddhartha Engineering College

- Led a student club promoting electric-vehicle awareness; organised technical events, workshops and industry collaborations.

Intern — Power Electronics Jun - Jul 2024 (6 weeks)

National Institute of Technology, Andhra Pradesh (NIT AP)

- Implemented and compared **MPPT** algorithms for solar PV systems, modelling converter behaviour and tracking efficiency in MATLAB/Simulink.

Internship Trainee — Cyber Security May - Jul 2024 (12 weeks)

SkillDzire — Virtual

- Completed a structured programme covering network security fundamentals, threat analysis and secure system practices.

Industrial Trainee Oct 2021 - Feb 2022 (16 weeks)

AP Central Power Distribution Corporation Ltd (APCPDCL)

- Trained at the 33/11 kV substation, Vuyyuru under an Executive Engineer, covering transformer operation, protection schemes and switchyard maintenance.

PUBLICATIONS

Enhancing Mobility in Autonomous Robotics with a Four-Wheel, Four-Steering System Controlled Remotely via Ubuntu

Accepted, IEEE BITCON 2024 — four-wheel independent steering architecture for improved manoeuvrability of autonomous field robots.

Precision Chili Harvesting with ROS 2 and Reinforcement Learning

Accepted, IEEE BITCON 2024 — integration of ROS 2 with reinforcement learning to automate selective harvesting of chili crops.

Portable Solar Charging System for Electric Bicycles and Batteries

Accepted, IEEE BITCON 2024 — solar-based charging solution for electric bicycles and portable battery packs.

TECHNICAL SKILLS

Hardware & PCB: KiCad, Altium Designer, schematic capture, multi-layer PCB layout & routing, component selection, circuit design, hardware bring-up and debug

Robotics: ROS / ROS 2, Ignition Fortress (Gazebo), swerve-drive control, sensor integration, mini-bot / robot platform design

Power & Energy: Power electronics, MPPT for solar, renewable energy systems, power systems & distribution

Programming: Python, MicroPython, Embedded C

Software Tools: MATLAB, Simulink, PSpice, MiPower, PowerWorld, Thonny, MS Office

Platforms: Linux (Ubuntu), Windows, macOS

Professional: Team leadership, problem solving, adaptability, critical thinking, time management

EDUCATION

B.Tech, Electrical and Electronics Engineering — CGPA 8.37 Sep 2022 - Apr 2025

Velagapudi Ramakrishna Siddhartha Engineering College, Vijayawada (JNTUK)

Diploma, Electrical and Electronics Engineering — 90.09% Jul 2019 - Apr 2022

Government Polytechnic, Gannavaram (SBTET, Andhra Pradesh)

Secondary Education — CGPA 9.5 Jul 2014 - Apr 2019

Z.P. High School, Bhatlapenumarru

CERTIFICATIONS & AWARDS

- Cisco Certified Network Professional (CCNP) — Cisco
- NDG Linux Unhatched & Basic Introduction to Linux — Cisco Networking Academy
- Overview of Space Science and Technology — ISRO
- Introduction to Internet of Things — NPTEL (Score: 80/100)
- The Joy of Computing using Python; Privacy & Security in Online Social Media — NPTEL
- Solder Master — 2nd Prize**, AFOSEC Event, VR Siddhartha Engineering College (Feb 2023)
- Participant, AP State Energy Conservation Mission (APSECM) Rally (Dec 2022)